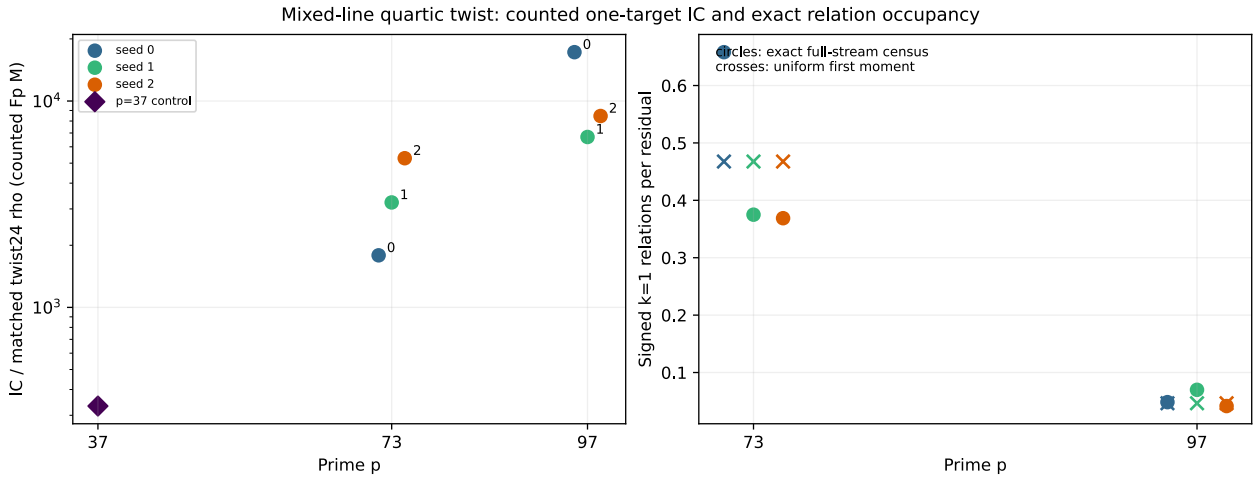
# Twist symmetry in mixed-component polynomial index calculus

23 October 2026. Public synthetic one-target quartic-twist workloads at $p = 73$ and $p = 97$, each with $Q = [271]G$ and three frozen independent seeds. Protocol, raw records, independent audits, per-run PDFs and source links are in `K1_SCALING_REPORT.md`, `K1_SCALE_RUNS.md`, and `K1_CENSUS_RUNS.md`.

## Mechanism

The transported Frobenius and cubic automorphism give a signed 24-action. Each tested nonzero abscissa orbit has size 12. A mixed base with $B_0$ points on $L_0 = F_p$ and $B_2$ on $L_2 = t^2 F_p$ has $\frac{B_0+B_2}{12}$ folded log columns: eight for the $48 + 48$ base at $p = 73$, six for the $36 + 36$ base at $p = 97$. Restricting to three $L_0$ factors and one $L_2$ factor permits a signed $L_2$ peel followed by a symmetrised $S_4$ solve in three $L_0$ abscissae. The 24 images of a folded relation row are scalar multiples of that row, so symmetry does not provide 24 new independent equations. This was checked on all 42 selected rows under all 24 actions: 1,008 exact row/RHS equalities and 12,096 factor-image closure checks.




## Six one-target polynomial runs

| p/seed | Queries | Exact/poly | Rank | IC Fp M | IC/rho Fp M |
|---|---|---|---|---|---|
| 73/0 | 15 | 8/8 | 8/8 | 888,203,272 | 1,788 |
| 73/1 | 20 | 11/9 | 8/8 | 1,183,617,228 | 3,225 |
| 73/2 | 43 | 15/15 | 8/8 | 2,542,727,892 | 5,283 |
| 97/0 | 310 | 16/16 | 6/6 | 13,839,508,536 | 17,249 |
| 97/1 | 75 | 6/6 | 6/6 | 3,349,559,070 | 6,695 |
| 97/2 | 150 | 6/6 | 6/6 | 6,696,848,851 | 8,461 |

Every target was independently verified as 271. Five processed-prefix signed sets matched exactly. The $p = 73$, seed-one $S_4$ collector missed two relations at one failed system (query index 18), but still reached anchored rank eight and recovered the target. It remains an explicit partial relation-finder result. All first/A/A non-timing records match.

## Complete frozen-stream occupancy

There are exactly $16\binom{B_0}{3}B_2$ signed distinct-index $k = 1$ tuples, giving expected set size $\lambda_1 = 16\binom{B_0}{3}\frac{B_2}{r}$ for a uniformly random residual. This first moment does not imply independent hits or guaranteed matrix rank. A point-keyed group census and independent BSGS-derived scalar-keyed auditor matched all 1,856 per-query complete-set digests, including post-stop queries.

| p/seed | Stream | Exact k=1 | Zero queries | Uniform total |
|---|---|---|---|---|
| 73/0 | 117 | 77 | 57 | 54.71 |
| 73/1 | 120 | 45 | 80 | 56.11 |
| 73/2 | 122 | 45 | 88 | 57.05 |
| 97/0 | 558 | 27 | 535 | 25.92 |
| 97/1 | 458 | 32 | 427 | 21.27 |
| 97/2 | 481 | 20 | 461 | 22.34 |

At $p = 97$, 1,423 of 1,497 residuals have no $k = 1$ relation. The polynomial producer agrees with the exact oracle on every processed $p = 97$ query, so low physical occupancy explains these long collection paths. A one-count mutation in the census was rejected by the scalar auditor at query zero.

## Claim boundary

The displayed ratios compare counted field multiplications only. Native hash/tree and allocator CPU, requested peak bytes, calibrated group/field bands, isolated interleaved timing, and same-input msolve/Magma results remain missing. They are not whole-method speed ratios or an admitted leaderboard result. The p=37 control had a counted ratio of 332.6 on a different base. This finite three-prime panel does not establish an asymptotic trend or a deployed-curve claim. The complete Markdown report names the next failure repair, base-shape comparison, and pricing gates.